

## LYCKSELE, Sweden

WMO: 022610

Lat: 64.5481N

Lon: 18.7130E

Elev: 211

StdP: 98.82

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|-------------|------------|-------------|-----------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |             | 1%         |             |                       |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-31.3 | (c)<br>-28.0 | (d)<br>-34.5                  | (e)<br>0.1 | (f)<br>-31.3 | (g)<br>-31.0 | (h)<br>0.2 | (i)<br>-27.9 | (j)<br>8.4            | (k)<br>-3.3 | (l)<br>7.5 | (m)<br>-3.6 | (n)<br>0.3            | (o)<br>310 | (p)<br>0.672 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.4                   | 26.4            | 16.3 | 24.3 | 15.8 | 22.3 | 14.8 | 18.1                | 23.3 | 17.0 | 21.8 | 16.0 | 20.4 | 3.3                  | 150  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.4                            | 12.0 | 19.5 | 15.2 | 11.1 | 18.3 | 14.1 | 10.3 | 17.2 | 52.0          | 23.6 | 48.5 | 21.8 | 45.4 | 20.4 | 22.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |      |  |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|------|--|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |      |  |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      |      |  |
| (a) 7.9           | (b) 6.7 | (c) 5.9 | (e) -34.6                  | (f) 29.2 | (g) 3.2            | (h) 2.2 | (i) -36.9                                          | (j) 30.7 | (k) -38.7  | (l) 32.0 | (m) -40.5  | (n) 33.3 | (o) -42.9  | (p) 34.9 |      |  |
|                   |         |         | DB                         |          |                    |         |                                                    |          |            |          |            |          |            |          |      |  |
|                   |         |         | WB                         | -34.7    | 19.5               | 3.2     | 1.4                                                | -37.0    | 20.6       | -38.8    | 21.4       | -40.6    | 22.2       | -42.9    | 23.3 |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 1.8    | -11.2 | -9.7  | -5.0  | 1.3   | 7.3   | 11.9  | 15.2  | 13.1  | 8.3   | 1.7   | -4.1  | -8.1  |
|                                                                        | DBStd       | 10.48  | 8.22  | 8.03  | 6.18  | 3.77  | 3.93  | 3.43  | 3.01  | 3.19  | 3.43  | 4.26  | 5.96  | 7.75  |
|                                                                        | HDD10.0     | 3378   | 658   | 552   | 464   | 261   | 103   | 18    | 1     | 8     | 72    | 257   | 423   | 561   |
|                                                                        | HDD18.3     | 6051   | 917   | 786   | 723   | 511   | 342   | 195   | 105   | 166   | 301   | 515   | 673   | 819   |
|                                                                        | CDD10.0     | 382    | 0     | 0     | 0     | 0     | 19    | 74    | 163   | 103   | 21    | 1     | 0     | 0     |
|                                                                        | CDD18.3     | 14     | 0     | 0     | 0     | 0     | 0     | 2     | 9     | 3     | 0     | 0     | 0     | 0     |
|                                                                        | CDH23.3     | 284    | 0     | 0     | 0     | 0     | 14    | 58    | 158   | 53    | 0     | 0     | 0     | 0     |
|                                                                        | CDH26.7     | 48     | 0     | 0     | 0     | 0     | 1     | 12    | 31    | 4     | 0     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 2.3    | 1.8   | 2.0   | 2.5   | 2.5   | 2.9   | 3.0   | 2.5   | 2.2   | 2.3   | 2.1   | 1.9   | 1.9   |
| (15) Precipitation                                                     | PrecAvg     | 600    | 38    | 26    | 28    | 29    | 43    | 67    | 96    | 81    | 55    | 48    | 45    | 41    |
|                                                                        | PrecMax     | 784    | 79    | 66    | 54    | 52    | 77    | 105   | 163   | 200   | 138   | 106   | 93    | 90    |
|                                                                        | PrecMin     | 442    | 4     | 5     | 11    | 6     | 8     | 33    | 35    | 22    | 12    | 17    | 15    | 22    |
|                                                                        | PrecStd     | 89     | 15    | 14    | 12    | 15    | 20    | 23    | 36    | 45    | 30    | 22    | 24    | 17    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 6.0   | 6.9   | 9.3   | 18.0  | 25.3  | 28.4  | 30.0  | 27.1  | 20.5  | 13.8  | 9.4   | 5.8   |
|                                                                        |             | MCWB   | 4.0   | 4.2   | 5.0   | 9.5   | 13.7  | 15.7  | 18.4  | 17.2  | 14.4  | 10.2  | 7.9   | 3.8   |
|                                                                        | 2%          | DB     | 3.8   | 5.1   | 7.0   | 13.6  | 22.2  | 24.9  | 26.9  | 24.9  | 18.1  | 11.1  | 6.5   | 4.2   |
|                                                                        |             | MCWB   | 2.2   | 2.8   | 3.4   | 7.2   | 12.7  | 14.9  | 16.6  | 16.6  | 13.2  | 8.8   | 5.3   | 2.6   |
|                                                                        | 5%          | DB     | 1.7   | 3.1   | 5.5   | 10.8  | 19.3  | 22.0  | 25.0  | 22.6  | 16.0  | 9.6   | 4.5   | 2.8   |
|                                                                        |             | MCWB   | 0.6   | 1.3   | 2.4   | 5.6   | 11.4  | 13.5  | 16.1  | 15.9  | 12.1  | 7.9   | 3.4   | 1.7   |
|                                                                        | 10%         | DB     | 0.1   | 1.0   | 4.0   | 8.5   | 16.2  | 19.8  | 22.9  | 20.1  | 14.4  | 8.2   | 3.0   | 1.4   |
|                                                                        |             | MCWB   | -0.4  | 0.0   | 1.4   | 4.1   | 9.6   | 12.4  | 15.5  | 14.9  | 11.5  | 6.9   | 2.3   | 0.7   |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 3.9   | 4.2   | 5.2   | 9.8   | 15.4  | 17.5  | 20.3  | 19.3  | 15.3  | 11.2  | 8.1   | 4.1   |
|                                                                        |             | MCDB   | 5.8   | 6.5   | 8.5   | 16.8  | 21.6  | 24.8  | 24.8  | 24.2  | 18.2  | 12.7  | 9.3   | 5.5   |
|                                                                        | 2%          | WB     | 2.3   | 3.0   | 3.8   | 7.4   | 13.5  | 15.7  | 18.6  | 17.7  | 14.1  | 9.3   | 5.5   | 2.8   |
|                                                                        |             | MCDB   | 3.6   | 5.1   | 6.6   | 12.3  | 20.8  | 22.2  | 23.5  | 22.3  | 16.7  | 10.7  | 6.4   | 3.8   |
|                                                                        | 5%          | WB     | 0.7   | 1.4   | 2.7   | 5.8   | 11.9  | 14.5  | 17.6  | 16.7  | 13.1  | 8.1   | 3.6   | 1.7   |
|                                                                        |             | MCDB   | 1.3   | 2.8   | 5.2   | 10.0  | 18.6  | 20.0  | 22.7  | 20.8  | 15.3  | 9.4   | 4.3   | 2.4   |
|                                                                        | 10%         | WB     | -0.5  | -0.1  | 1.5   | 4.4   | 10.1  | 13.3  | 16.5  | 15.5  | 12.1  | 6.9   | 2.3   | 0.7   |
|                                                                        |             | MCDB   | 0.1   | 0.7   | 3.6   | 8.0   | 15.2  | 18.1  | 21.1  | 19.2  | 13.9  | 8.1   | 2.9   | 1.3   |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 9.2   | 10.3  | 11.4  | 11.0  | 12.2  | 11.3  | 11.4  | 11.1  | 10.0  | 7.5   | 6.5   | 8.4   |
|                                                                        | 5% DB       | MCDBR  | 8.9   | 9.6   | 11.3  | 15.6  | 18.3  | 15.8  | 15.6  | 16.0  | 12.6  | 9.7   | 6.8   | 8.5   |
|                                                                        |             | MCWBR  | 8.2   | 8.1   | 8.2   | 9.4   | 9.7   | 6.9   | 6.7   | 8.4   | 8.2   | 7.3   | 6.0   | 7.9   |
|                                                                        | 5% WB       | MCDBR  | 8.6   | 9.0   | 10.5  | 14.2  | 17.2  | 13.6  | 13.2  | 12.7  | 10.9  | 8.8   | 6.5   | 8.0   |
|                                                                        |             | MCWBR  | 8.0   | 7.8   | 7.9   | 9.0   | 9.5   | 6.5   | 6.6   | 7.2   | 7.4   | 7.0   | 5.8   | 7.5   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.194 | 0.267 | 0.297 | 0.331 | 0.337 | 0.338 | 0.350 | 0.338 | 0.308 | 0.291 | 0.336 | 0.352 |
|                                                                        | taud        |        | 1.971 | 2.186 | 2.248 | 2.338 | 2.467 | 2.491 | 2.471 | 2.496 | 2.565 | 2.440 | 2.695 | 2.735 |
|                                                                        | Ebn at Noon |        | 361   | 641   | 786   | 843   | 874   | 879   | 855   | 833   | 792   | 629   | 343   | 177   |
|                                                                        | Edh at Noon |        | 24    | 57    | 83    | 97    | 94    | 94    | 93    | 83    | 63    | 45    | 21    | 14    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 0.14  | 0.77  | 2.30  | 3.99  | 4.75  | 5.24  | 4.84  | 3.68  | 2.19  | 0.89  | 0.21  | 0.05  |
|                                                                        | RadStd      |        | 0.01  | 0.12  | 0.20  | 0.29  | 0.58  | 0.44  | 0.56  | 0.36  | 0.27  | 0.15  | 0.04  | 0.00  |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (40 neighbors) |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page